Individual Comparisons of Elements in a Set as a Method of Ranking

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May 6, 2026

Preface

A document detailing thoughts on how one might design an experiment reviewing the effectiveness of "which is better" style surveys, and how their structures could create biases in the data.

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Premise

In this document we explore the “which do you prefer” style of survey used for ranking a list of things. Suppose we have a list of things that we want a participant in our survey to rank with respect to some subjective criterion (e.g. favourite to least favourite, best to worst). If the list of potential things to rank is overwhelmingly large, it is very difficult for a participant to accurately rank all of them manually.

However, if we instead present our participant two elements from the list, and ask which of the two they would rank above the other based on the subjective criterion, the survey becomes far more tractible. This type of survey is what we will explore. We will refer to the “list of things” as a set of objects, and to each individual comparison we ask the participant to make as a question.

Chapter 1

Building a Mathematical Model

We will seek to construct a mathematical model with which we can analyse our topic of interest more abstractly.

In this chapter, we may assume

$$i, j, k, n \in \mathbb{N}, \text{ and } i, j \leq n$$

1.1 Objects, Questions, and Rankings

To begin, our model should include a list of objects to be ranked, and some criterion with respect to which the objects are ranked. Let us define the following to begin constructing this model.

Definition 1.1.1. List / Set of Objects: Let $L = \{l_1, l_2, \cdots, l_n\}$ be a set of n objects. Then L is our **list of objects** or, equivalently, our **set of objects**.

Definition 1.1.2. Ranking: Let $r_i, \in \mathbb{N}$ be a natural number. Then r_i is the **ranking** of l_i if

$$\forall i \in \mathbb{N}, i \leq n, \exists!\ r_i \in \mathbb{N} : r_i \leq n$$

Definition 1.1.3. Relative Ranking: Let $\leq : L \times L \longrightarrow \{\text{True, False}\}$ be a binary relation on L. Then we have

$$l_i \leq l_j \iff r_i \leq r_j \iff l_i \text{ ranks lower than } l_j$$

Definition 1.1.4. Questions: Let $q_{ij} = q_{ji} = \{l_i, l_j\} \subseteq L$ be the unordered pair containing the i^{th} and j^{th} elements in the set L. Then q_{ij} also denotes the **question** asked to a participant that is of the form

$$\text{“Which is true: } l_i \leq l_j \text{ or } l_j \leq l_i\text{?”}$$

Remark 1.1.5. We will also let $Q = \{q_{ij} \mid i \neq j,\ \forall i, j \leq n\}$ the set of all questions we could ask a participant, and let $q_k = \{q_{ij} \in Q \mid i = k \text{ or } j = k\}$ all questions involving l_k.

This model is sound in theory, but flawed in practice. While it works for any well defined binary option $\leq$, it doesn't account for subjective criteria, making it unsuitable for an actual survey. Therefore, it would be more useful to instead have each element in the list be given a score based on how many times it was picked over another during the survey.

1.2 Statistics and Participants

1.3 An Individual Participant

Let P_i denote participant with the identifier i.

Consider a single participant, P_1, that takes the survey, and suppose they are willing to answer arbitrarily many questions.

i) What is the minimum number q_1 of total questions given to participant 1 required such that every element l_i in the set L has been ranked based on answers to at least $N_p(l_i)$ questions given to participant 1?

ii) How should the rankings $R(l_i)$ be determined from the answers given to questions?

iii) How would one merge rankings into an average to see the average ranking for the list

iv) Should some questions be shown first over others when surveying multiple people to account for potential biases?

v) How does the phrasing of the criterion affect the outcome of the survey?

Chapter 2

Formalizing our Mathematical Model